Torsional friction in planar contact

Sébastien Kleff
sk8001@nyu.edu

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Abstract

Tactile sensors are widely used to detect slip, but much less is known about what can actually be inferred about frictional contact limits from tactile measurements. This question matters because detecting that slip is occurring or imminent is not the same as knowing which tangential and torsional loads the contact can still support. That distinction is important for control design. In this report, we study this question for a single planar contact patch under quasi-static Coulomb friction. We first show that coarse normal force information, consisting of the total normal force and center of pressure, is generally insufficient to determine the admissible tangential wrench set: two contacts can have the same total normal force and CoP while supporting different combinations of tangential force and torsional moment. We then study what can still be inferred from partial measurements. In particular, we derive upper and lower bounds on the admissible wrench set from coarse measurements and from spatially distributed taxel-wise normal forces. The latter provide tighter bounds by reducing the ambiguity on how the normal force is distributed over the contact patch. Taking the torques about a fixed point makes every bound a generalized moment problem: the bounds from taxel forces have closed forms, those combining taxel forces and CoP are an SOCP and an LP, and the gap between them halves, to first order, with each dyadic refinement of the taxel grid. Along a given load, the guaranteed and the possible wrench sets give a certified and an upper load factor, each a convex program, which decide whether the load is surely carried, surely not carried, or undecided by the measurement.

1 Introduction

Modern tactile sensors provide rich local information about contact pressure, shape, and deformation. Yet tactile features used for control are often selected mainly from empirical performance, for example contact pose, resultant force, image features, or local pressure measurements. A more basic question is less often made explicit: given a particular tactile measurement, what can actually be inferred about the physical interaction at the contact? We study this question in the setting of a rigid planar contact patch under Coulomb friction. Rather than trying to characterize every possible tactile signal or manipulation task, we focus on one concrete control-relevant quantity: the set of tangential forces and torsional moments that the contact can support without violating the friction model. The key difficulty is that this admissible wrench set depends on how the normal load is distributed over the patch. Coarse measurements such as the total normal force and center of pressure constrain this load distribution, but do not determine it uniquely. Spatial tactile measurements provide additional information and therefore reduce this ambiguity. The objective of this report is to make this relationship explicit. We ask **how the admissible frictional wrench set can be bounded from different levels of normal force information**, from total force and CoP to spatially distributed taxel-wise forces, **without requiring the full normal pressure field**.

2 Preliminaries (measures)

We briefly introduce the measure-theoretic notation used in the following. Only the notions needed for the contact model are recalled here; standard reference is [1].

2.1 Force as a measure.

Let $P \subset \mathbb{R}^2$ denote the contact patch, equipped with its Borel σ-algebra $\mathcal{B}(P)$. A *nonnegative measure* $\nu_N : \mathcal{B}(P) \to [0, +\infty]$ assigns to every measurable subset $A \subseteq P$ a nonnegative scalar $\nu_N(A)$, and is countably additive, i.e. for any pairwise disjoint measurable sets $A_i \subseteq P$,

$$\nu_N\left(\bigcup_i A_i\right) = \sum_i \nu_N(A_i). \tag{1}$$

In the present context, ν_N represents the (spatially distributed) normal force, i.e. $\nu_N(A)$ is the normal force carried by the region A. Hence $\nu_N(A)$ has units of force (N), and the resultant normal force is $f_N = \nu_N(P)$. We restrict attention to *finite nonnegative Radon measures*, denoted $\nu_N \in \mathbb{M}_+(P)$. Informally, a Radon measure is a "well-behaved" measure. In particular, on the compact sets considered here, it is finite and its value on measurable sets can be approximated from compact subsets. For our purpose, it essentially means that our measure does not assign an infinite force to P and it can represent either the force induced by an ordinary pressure field or an idealized concentrated point/line load.

We represent the tangential force by an $\mathbb{R}^2$-valued finite Radon measure $\nu_T = (\nu_{T,1}, \nu_{T,2})$ where each component $\nu_{T,i}$ is a finite signed Radon measure on P. Thus, for every measurable region $E \subseteq P$, the resultant tangential force carried by E is

$$\nu_T(E) = \begin{bmatrix} \nu_{T,1}(E) \\ \nu_{T,2}(E) \end{bmatrix} \in \mathbb{R}^2, \tag{2}$$

2.2 Area measure and pressure fields.

Let dA denote the usual area element on the planar patch P. A *normal pressure field* is a nonnegative function $\rho_N : P \to \mathbb{R}_+$ with units of force per unit area. If $\rho_N \in L^1_+(P)$, which simply means that ρ_N has finite integral over P

$$\rho_N(x) \geq 0 \quad \text{a.e.}, \qquad \int_P \rho_N(x)\, dA < \infty. \tag{3}$$

then it defines a finite normal force measure: the force carried by any region $E \subseteq P$ is given by

$$\nu_N(E) = \int_E \rho_N(x)\, dA. \tag{4}$$

which we also write equivalently as $d\nu_N(x) = \rho_N(x)\, dA$. In that case, ρ_N is called the *density* of the normal force measure ν_N with respect to area[1]. Not every normal force measure admits such a pressure field. For example the Dirac measure $\nu_N = f_N \delta_c$ represents an idealized normal force f_N concentrated at the single point c. It cannot be represented by an ordinary pressure field over area. In this report we adopt the measure-theoretic language (instead of only L^1_+ densities) so that we cover both "classical" pressure fields and idealized concentrated loads.

This notation allows ordinary pressure fields and concentrated loads to be handled by the same integration equations. For a measurable scalar or vector function ψ, the integral

$$\int_P \psi(x)\, d\nu_N(x) \tag{5}$$

[1] More generally, the Radon–Nikodym theorem guarantees the existence of a density for measures under suitable assumptions, such as "absolute continuity" which means physically that a subset of zero area (e.g. a point or a line) carries zero normal force. See [1], Theorem 1.30, p. 50

can be interpreted as a force-weighted spatial integral. Whenever ν_N admits a normal pressure field ρ_N,

$$\int_P \psi(x)\, d\nu_N(x) = \int_P \psi(x)\rho_N(x)\, dA. \tag{6}$$

For a Dirac measure,

$$\int_P \psi(x)\, d(f_N\delta_c)(x) = f_N\psi(c). \tag{7}$$

2.3 Moments of the normal force measure.

The zeroth moment of ν_N is its total measure,

$$m_0(\nu_N) \triangleq \int_P 1\, d\nu_N = \nu_N(P) = f_N. \tag{8}$$

In the present mechanical interpretation, this is the *resultant normal force*. The first moment is

$$m_1(\nu_N) \triangleq \int_P x\, d\nu_N(x) \in \mathbb{R}^2. \tag{9}$$

If $f_N > 0$, the *center of pressure* is the normalized first moment,

$$c[\nu_N] \triangleq \frac{m_1(\nu_N)}{m_0(\nu_N)} \tag{10}$$

$$= \frac{1}{f_N}\int_P x\, d\nu_N(x). \tag{11}$$

Hence knowing the total normal force and the center of pressure is equivalent to knowing the zeroth and first moments of the normal force measure $m_0(\nu_N) = f_N, m_1(\nu_N) = f_N c[\nu_N]$. Unlike the CoP, the pair (m_0, m_1) is linear in ν_N and defined for every measure, including $\nu_N = 0$. We work with these *homogeneous coordinates* throughout, and use the CoP only when $f_N > 0$.

3 Problem formulation

3.1 Planar contact model

Frame. All quantities are coordinates in a fixed frame $\mathcal{R} = (O, \mathcal{B})$ of the contact plane: O is the origin and $\mathcal{B} = (e_1, e_2)$ an orthonormal basis, completed by the contact plane normal e_3. A point of the plane is identified with its coordinate vector $x \in \mathbb{R}^2$. A wrench or a twist is always written in a frame, and its coordinates depend on the reference point (Lemma 3.4); here the reference point is O unless stated otherwise. For $x, y \in \mathbb{R}^2$, the *planar cross product* is the scalar

$$x \times y \triangleq \langle e_3, (x,0)\times(y,0)\rangle = x_1y_2 - x_2y_1 = (Jx)^\top y, \qquad J \triangleq \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}. \tag{12}$$

Let the planar contact patch $P \subset \mathbb{R}^2$ be compact, and let $\nu_N \in \mathbb{M}_+(P)$ and ν_T be the normal and tangential force measures introduced above. We further assume a known and spatially uniform Coulomb friction coefficient $\mu > 0$. We impose Coulomb friction in measure form by requiring that there exists a measurable density $\gamma_T : P \to \mathbb{R}^2$ of ν_T with respect to ν_N such that

$$d\nu_T(x) = \gamma_T(x)\, d\nu_N(x), \qquad \|\gamma_T(x)\|_2 \le \mu \quad \nu_N\text{-a.e. on } P. \tag{13}$$

We call γ_T the *friction ratio field*. It can be interpreted as the tangential force per unit normal force. The above condition expresses both that tangential friction force can only be carried where

normal force is present and that its local magnitude is bounded by μ[2]. When ν_N admits the normal pressure field ρ_N introduced above, i.e. $d\nu_N = \rho_N\, dA$, the tangential force measure also admits a density with respect to area,

$$\rho_T(x) = \gamma_T(x)\rho_N(x), \tag{14}$$

called the *tangential traction field*. The measure-form Coulomb condition then reduces to the classical local condition

$$\|\rho_T(x)\|_2 \le \mu\rho_N(x) \qquad \text{a.e. on } P. \tag{15}$$

We define the *resultant tangential force* and *resultant torsional moment* about the origin O as

$$f_T \triangleq \nu_T(P) = \int_P \gamma_T(x)\, d\nu_N(x), \qquad \tau_N \triangleq \int_P x \times \gamma_T(x)\, d\nu_N(x). \tag{16}$$

Definition 3.1 (Admissible tangential wrench set)**.** *For a normal force measure $\nu_N \in \mathbb{M}_+(P)$, the admissible tangential wrench set is*

$$\mathcal{X}(\nu_N) \triangleq \left\{ (f_T, \tau_N) \in \mathbb{R}^2 \times \mathbb{R} : \begin{array}{l} \exists\, \gamma_T : P \to \mathbb{R}^2 \text{ measurable such that} \\ \|\gamma_T(x)\|_2 \le \mu \quad \nu_N\text{-a.e.}, \\ f_T = \displaystyle\int_P \gamma_T(x)\, d\nu_N(x), \\ \tau_N = \displaystyle\int_P x \times \gamma_T(x)\, d\nu_N(x) \end{array} \right\}. \tag{17}$$

It is a nonempty, convex, compact subset of $\mathbb{R}^3$, symmetric about 0, containing all resultant tangential wrenches (tangential force and torsional moment about O) that are compatible with the normal force measure ν_N and the local Coulomb friction condition (13).

Proposition 3.2 (Support function of the admissible tangential wrench set)**.** *For any $\nu_N \in \mathbb{M}_+(P)$ and for every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$, the support function of $\mathcal{X}(\nu_N)$ is*

$$h_{\mathcal{X}(\nu_N)}(q) = \mu \int_P \|a + bJx\|_2 \; d\nu_N(x). \tag{18}$$

Proof. By definition, $\forall q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$, $h_{\mathcal{X}(\nu_N)}(q) = \sup_{(f_T,\tau_N)\in\mathcal{X}(\nu_N)} \left(a^\top f_T + b\tau_N\right)$. For any friction ratio field γ_T,

$$a^\top f_T + b\tau_N = \int_P [a + bJx]^\top \gamma_T(x)\, d\nu_N(x) \le \mu \int_P \|a + bJx\|_2 \; d\nu_N(x), \tag{19}$$

by Cauchy–Schwarz and $\|\gamma_T(x)\|_2 \le \mu$. The bound is attained by the field of magnitude μ aligned with $a + bJx$ where this vector is nonzero, and $\gamma_T^\star(x) = 0$ elsewhere. □

Remark 3.3 (Twists and dissipated power)**.** *The direction $q = (a, b)$ is the coordinate vector of a planar twist about O: a is the velocity of the material point at O and b the angular velocity, so the point at x has velocity $v_q(x) = a + bJx$. The power of the tangential forces under this twist is $\int_P \langle v_q(x), \gamma_T(x)\rangle\, d\nu_N(x) = a^\top f_T + b\tau_N = \langle q, (f_T, \tau_N)\rangle$. Hence $h_{\mathcal{X}(\nu_N)}(q)$ is the largest power that admissible friction can exert under the twist q; since $\mathcal{X}(\nu_N) = -\mathcal{X}(\nu_N)$, it is also the maximal power that friction can dissipate when the contact slides with twist q, which is the value selected by the maximal dissipation principle.*

[2]In particular, $d\nu_T = \gamma_T\, d\nu_N$ implies $\nu_T \ll \nu_N$. Conversely, by the Radon–Nikodym theorem, a tangential measure with $\|\nu_T(E)\|_2 \le \mu\, \nu_N(E)$ for every E has such a density.

Symbol	Meaning	Remark	Lives in
ν_N	normal force measure	true $\nu_N^\star$ is unknown	$\mathbb{M}_+(P)$
(m_0, m_1)	zeroth and first moments of ν_N	$m_1 = f_N c[\nu_N]$ if $f_N > 0$	$\mathbb{R} \times \mathbb{R}^2$
$q = (a, b)$	twist about O (support direction)	prescribed	$\mathbb{R}^2 \times \mathbb{R}$
$\mathcal{M}$	measurement operator	fixed by the sensing model	$\mathbb{M}_+(P) \to \mathbb{R}^r$
y	measurement value	observed data from $(\mathcal{M}, \nu_N)$	$\mathbb{R}^r$
$\Pi_{\mathcal{M}}(y)$	all measures compatible with y	depends on $(\mathcal{M}, y)$	$\subseteq \mathbb{M}_+(P)$
$\underline{h}(q; y, \mathcal{M})$	measurement-conditioned lower bound	depends on $(\mathcal{M}, y, q)$	$\mathbb{R}$
$\overline{h}(q; y, \mathcal{M})$	measurement-conditioned upper bound	depends on $(\mathcal{M}, y, q)$	$\mathbb{R}$
$\mathcal{X}^-(y, \mathcal{M})$, $\mathcal{X}^+(y, \mathcal{M})$	guaranteed and possible wrench sets	Proposition 3.10	$\subseteq \mathbb{R}^3$

Table 1: Notations.

Lemma 3.4 (Change of reference point)**.** *For $o \in \mathbb{R}^2$, let $T_o(f, \tau) \triangleq (f, \tau - o \times f)$ and $S_o(a, b) \triangleq (a + bJo, b)$ (wrench and twist coordinates about o; Varignon's formula). Then T_o and S_o are invertible linear maps and*

(i) $\langle S_o q, T_o w \rangle = \langle q, w \rangle$ for all $q, w \in \mathbb{R}^3$: the power does not depend on the reference point;

(ii) the set of wrenches about o (over the same friction ratio fields γ_T),

$$\mathcal{X}_o(\nu_N) \triangleq \left\{ \left(f_T, \textstyle\int_P (x - o) \times \gamma_T \, d\nu_N \right) \right\},$$

is $T_o \mathcal{X}(\nu_N)$, and $h_{\mathcal{X}_o(\nu_N)}(S_o q) = h_{\mathcal{X}(\nu_N)}(q)$, i.e.,

$$h_{\mathcal{X}_o(\nu_N)}(a', b) = \mu \int_P \|a' + bJ(x - o)\|_2 \, d\nu_N(x).$$

Proof. (i) $\langle S_o q, T_o w \rangle = (a + bJo)^\top f + b(\tau - o \times f) = a^\top f + b\tau$, since $o \times f = (Jo)^\top f$. (ii) $\int_P (x - o) \times \gamma_T \, d\nu_N = \tau_N - o \times f_T$. Then $h_{\mathcal{X}_o(\nu_N)}(S_o q) = \sup_{w \in \mathcal{X}(\nu_N)} \langle S_o q, T_o w \rangle = h_{\mathcal{X}(\nu_N)}(q)$ by (i), and $S_o^{-1}(a', b) = (a' - bJo, b)$ gives the integral form. □

Lemma 3.5 (Linearity in the normal force)**.** *For $\nu_1, \nu_2 \in \mathbb{M}_+(P)$ and $t \geq 0$, $h_{\mathcal{X}(\nu_1 + \nu_2)} = h_{\mathcal{X}(\nu_1)} + h_{\mathcal{X}(\nu_2)}$ and $h_{\mathcal{X}(t\nu_1)} = t \, h_{\mathcal{X}(\nu_1)}$. Consequently $\mathcal{X}(\nu_1 + \nu_2) = \mathcal{X}(\nu_1) \oplus \mathcal{X}(\nu_2)$ (Minkowski sum), $\mathcal{X}(t\nu_1) = t \, \mathcal{X}(\nu_1)$ and $\mathcal{X}(0) = \{0\}$.*

Proof. The first claims follow from Proposition 3.2. The support function of a Minkowski sum is the sum of the support functions, and nonempty compact convex sets with the same support function are equal [4, Section 13]. □

3.2 Measurement-conditioned support bounds

We focus here on the case where we do not know the *true* force measure $\nu_N^\star$ but we have only access to some measurements of it (e.g. resultant normal force, CoP, etc.). The goal is to investigate what can be known about the support of $\mathcal{X}(\nu_N^\star)$ given partial measurements of $\nu_N^\star$.

Definition 3.6 (Measurement operator)**.** *A measurement operator is a map $\mathcal{M} : \mathbb{M}_+(P) \to \mathbb{R}^r$ of the form*

$$\mathcal{M}(\nu_N) = \begin{bmatrix} \int_P \phi_1(x) \, d\nu_N(x) \\ \vdots \\ \int_P \phi_r(x) \, d\nu_N(x) \end{bmatrix}, \qquad \forall \nu_N \in \mathbb{M}_+(P), \tag{20}$$

where the prescribed bounded Borel functions $\phi_j : P \to \mathbb{R}$ are called the measurement functions. For example, $\phi = 1$ measures the resultant normal force f_N, $\phi(x) = x_1$ and $\phi(x) = x_2$ its first moment $m_1 = f_N c$ (CoP in homogeneous coordinates) In the following, we denote a measurement value (or observation) induced by the (unknown) true force measure $\nu_N^\star$ as $y = \mathcal{M}(\nu_N^\star) \in \mathbb{R}^r$.

Definition 3.7 (Measurement-compatible normal force family)**.** *Given a measurement operator $\mathcal{M} : \mathbb{M}_+(P) \to \mathbb{R}^r$ and the observation $y \in \mathbb{R}^r$, the set of normal force measures compatible with y is the preimage*

$$\Pi_{\mathcal{M}}(y) \triangleq \mathcal{M}^{-1}[\{y\}] = \{\nu_N \in \mathbb{M}_+(P) : \mathcal{M}(\nu_N) = y\} \tag{21}$$

It contains all the normal force measures that are indistinguishable under the measurement y

Given an observation y of the unknown true force measure $\nu_N^\star$, we can compute measurement-conditioned lower/upper bounds on the support function $h_{\mathcal{X}(\nu_N^\star)}$ by optimizing over $\Pi_{\mathcal{M}}(y)$.

Definition 3.8 (Measurement-conditioned support bounds)**.** *Let $\mathcal{M}$ be a measurement operator and $y \in \mathbb{R}^r$ an observation. For a twist $q \in \mathbb{R}^2 \times \mathbb{R}$, the measurement-conditioned bounds in direction q are*

$$\underline{h}(q; y, \mathcal{M}) \triangleq \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} h_{\mathcal{X}(\nu_N)}(q) = \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mu \int_P \|a + bJx\|_2 \, d\nu_N(x), \tag{22}$$

$$\overline{h}(q; y, \mathcal{M}) \triangleq \sup_{\nu_N \in \Pi_{\mathcal{M}}(y)} h_{\mathcal{X}(\nu_N)}(q) = \sup_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mu \int_P \|a + bJx\|_2 \, d\nu_N(x), \tag{23}$$

and the guaranteed and possible wrench sets are

$$\mathcal{X}^-(y, \mathcal{M}) \triangleq \bigcap_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mathcal{X}(\nu_N), \qquad \mathcal{X}^+(y, \mathcal{M}) \triangleq \bigcup_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mathcal{X}(\nu_N). \tag{24}$$

The objective of both problems in (22) is linear in ν_N, and the constraints $\int_P \phi_j \, d\nu_N = y_j$ are linear moment constraints: *the bounds are Generalized Moment Problems (GMP)* [2], for every measurement operator.

Assumption 3.9 (Mass control)**.** *There exists $\beta \in \mathbb{R}^r$ such that $\sum_j \beta_j \phi_j(x) \geq 1$ for every $x \in P$.*

Under Assumption 3.9, every $\nu_N \in \Pi_{\mathcal{M}}(y)$ satisfies $\nu_N(P) \leq \beta^\top y$, so the compatible family has bounded mass. This assumption keeps the upper bounds finite: without it, force could be hidden where all the measurement functions vanish. For example, consider $P = [0, L] \times [0, \ell]$, $r = 1$, $\phi(x) = x_1$ (so $\mathcal{M}(\nu_N) = \int_P x_1 \, d\nu_N$), and the observation $y = 0$. For every $t \geq 0$ and $x_2 \in [0, \ell]$, the measure $\nu_t = t \, \delta_{(0, x_2)}$ is compatible. Then take $q = (a, 0)$ with $a \neq 0$; the integrand is $\|a + bJx\|_2 = \|a\|_2$ everywhere, so $h_{\mathcal{X}(\nu_t)}(q) = \mu \|a\|_2 \, t$. Letting $t \to \infty$ gives $\overline{h}(q; 0, \mathcal{M}) = +\infty$. To prevent this pathology, all measurement operators studied below measure the resultant normal force (directly or as sum of the taxel forces) and satisfy Assumption 3.9.

Proposition 3.10 (Bounds and wrench sets)**.** *Let $\mathcal{M}$ (with Assumption 3.9) and $y \in \mathbb{R}^r$.*

(i) The possible set $\mathcal{X}^+(y, \mathcal{M})$ is convex, and its support function is $\overline{h}(\cdot; y, \mathcal{M})$.

(ii) The guaranteed set is $\mathcal{X}^-(y, \mathcal{M}) = \{w : \langle q, w \rangle \leq \underline{h}(q; y, \mathcal{M}) \ \forall q\}$. Its support function is the closed convex envelope of $\underline{h}(\cdot; y, \mathcal{M})$, which is in general strictly below $\underline{h}$.

(iii) $y \mapsto \overline{h}(q; y, \mathcal{M})$ is concave and $y \mapsto \underline{h}(q; y, \mathcal{M})$ is convex on the set of observations $\mathcal{M}(\mathbb{M}_+(P))$, and both are positively homogeneous.

Proof. (i) Mixing two compatible measures gives a compatible measure ($\Pi_{\mathcal{M}}(y)$ is convex), and $\mathcal{X}$ is linear in the measure (Lemma 3.5). So a mixture of two wrenches of $\mathcal{X}^+$ is a wrench of the mixed measure, hence belongs to $\mathcal{X}^+$. The support function of a union is the supremum of the support functions, which is $\overline{h}$. (ii) A wrench belongs to $\mathcal{X}(\nu_N)$ for every compatible ν_N if and only if $\langle q, w\rangle \leq h_{\mathcal{X}(\nu_N)}(q)$ for all q and all compatible ν_N, i.e., $\langle q, w\rangle \leq \underline{h}(q; y, \mathcal{M})$ for all q (a closed convex set is the intersection of its supporting half-spaces). The support function of the set defined by these inequalities is the closed convex envelope of $\underline{h}$ [4, Section 13]. (iii) Mixing measures compatible with y_1 and y_2 gives a measure compatible with the mixed observation, and $h_{\mathcal{X}(\nu_N)}(q)$ is linear in ν_N. Taking the supremum (resp. infimum) over the two measures gives concavity (resp. convexity). Positive homogeneity follows by scaling the measure. □

Remark 3.11 (How to use the bounds)**.** *A candidate tangential wrench w (about O) is carried by every compatible normal force measure, i.e., $w \in \mathcal{X}^-(y, \mathcal{M})$, if and only if $\langle q, w\rangle \leq \underline{h}(q; y, \mathcal{M})$ for all q. Conversely, if $\langle q, w\rangle > \overline{h}(q; y, \mathcal{M})$ for some q, then $w \notin \mathcal{X}^+(y, \mathcal{M})$: no compatible measure carries w, and the contact slips whatever the actual pressure distribution. By Proposition 3.10(i), the second test is a convex membership test. The sets are closed: a wrench with $\langle q, w\rangle = h_{\mathcal{X}(\nu_N)}(q)$ for some q lies on the boundary of $\mathcal{X}(\nu_N)$, where the contact is at the onset of sliding with twist q (maximal dissipation). A strict inequality, or a margin, certifies the interior. Section 9 turns both tests into convex programs along a load direction*[3]*.*

3.3 General structure of the bounds

For $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$, we write

$$g(x; q) \triangleq \|a + bJx\|_2 \,, \tag{25}$$

the speed of the material point at x under the twist q, so that, by Proposition 3.2, $h_{\mathcal{X}(\nu_N)}(q) = \mu \int_P g(x; q)\, d\nu_N(x)$. For $b \neq 0$, we can write $a + bJx = bJ(x - p)$ where $p \triangleq Ja/b$ is the instantaneous center of rotation (ICR), so that $g(x; q) = |b|\, \|x - p\|_2$ [4]. For $b = 0$, $g(x; q) = \|a\|_2$.

Lemma 3.12 (Properties of the integrand)**.** *For every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$:*

(i) $x \mapsto g(x; q)$ is convex on $\mathbb{R}^2$;

(ii) for all $x, x' \in \mathbb{R}^2$, $|g(x; q) - g(x'; q)| \leq |b|\, \|x - x'\|_2$;

(iii) for every $\nu_N \in \mathbb{M}_+(P)$, we have $h_{\mathcal{X}(\nu_N)}(q) \geq \mu\, \|m_0(\nu_N)\, a + bJm_1(\nu_N)\|_2$. For $f_N > 0$, it reads $h_{\mathcal{X}(\nu_N)}(q) \geq \mu f_N\, g(c[\nu_N]; q)$: the maximal dissipated power is at least that of the whole normal force concentrated at the CoP.

Proof. (i) $g(\cdot; q)$ is a norm composed with an affine map. (ii) By the reverse triangle inequality and $\|J\|_2 = 1$. (iii) By the triangle inequality for vector-valued integrals, $\int_P \|a + bJx\|_2\, d\nu_N \geq \|\int_P (a + bJx)\, d\nu_N\|_2 = \|m_0 a + bJm_1\|_2$. □

The bounds of two measurement operators can only be compared if their observations come from the same normal force measure. This leads to an order on the information carried by measurement operators, which we can formalize (Yann).

[3] Because all support functions are positively homogeneous in q, it is sufficient to consider one normalized representative of each nonzero ray.

[4] Note that this is the classical planar sliding form of the dissipation: the sliding speed of each contact point is its distance to the ICR times the angular rate [5, Eq. 7], and the limit surface is the maximum-power envelope of this quantity [6, §8].

Measurement	Observed data	Lower bound	Upper bound
$\mathcal{M}_0$ (Sec. 4)	$y_0 = (f_N, m_1)$	closed form	LP over the vertices of P
$\mathcal{M}_F$ (Sec. 6)	$y_F = (F_1, \dots, F_r)$	closed form: $\sum_i$ of minima over the taxels	closed form: $\sum_i$ of maxima over the taxel corners
$\mathcal{M}_T$ (Sec. 7)	$y_T = (y_0, y_F)$	SOCP	LP over the taxel corners

Table 2: Measurement operators and structure of the resulting bounds. All bounds are GMPs; the CoP constraint of $\mathcal{M}_T$ couples the taxels.

Definition 3.13 (Information preorder)**.** *A measurement operator $\mathcal{M}$ is at least as informative as $\mathcal{M}'$, written $\mathcal{M} \succeq \mathcal{M}'$, if*

$$\Pi_{\mathcal{M}}(\mathcal{M}(\nu_N)) \subseteq \Pi_{\mathcal{M}'}(\mathcal{M}'(\nu_N)) \qquad \text{for every } \nu_N \in \mathbb{M}_+(P), \tag{26}$$

i.e., every measure that $\mathcal{M}$ cannot distinguish from ν_N is also indistinguishable from ν_N under $\mathcal{M}'$. The concatenation $\mathcal{M} \vee \mathcal{M}' \triangleq (\mathcal{M}, \mathcal{M}')$ is the operator with measurement functions $(\phi_1, \dots, \phi_r, \phi'_1, \dots, \phi'_{r'})$.

Proposition 3.14 (Properties of the information preorder)**.** *(i) $\succeq$ is a preorder (reflexive and transitive). It is not antisymmetric, and not total.*

(ii) $\mathcal{M} \succeq \mathcal{M}'$ if and only if there is a matrix $A \in \mathbb{R}^{r' \times r}$ with $\phi'_j(x) = \sum_k A_{jk}\phi_k(x)$ for every $x \in P$ and every j, i.e., $\mathcal{M}' = A\,\mathcal{M}$.

(iii) $\mathcal{M} \vee \mathcal{M}' \succeq \mathcal{M}$ and $\mathcal{M} \vee \mathcal{M}' \succeq \mathcal{M}'$, with $\Pi_{\mathcal{M}\vee\mathcal{M}'}((y, y')) = \Pi_{\mathcal{M}}(y) \cap \Pi_{\mathcal{M}'}(y')$.

Proof. (i) Reflexivity and transitivity are inclusions of compatible families. The preorder is not antisymmetric: scaling all measurement functions by 2 gives a different operator with the same compatible families. It is not total: $\mathcal{M}_0$ and $\mathcal{M}_F$ of the next sections are incomparable (see (ii)). (ii) ($\Leftarrow$) If $\mathcal{M}' = A\mathcal{M}$, a measure with the same $\mathcal{M}$-value as ν_N has the same $\mathcal{M}'$-value. ($\Rightarrow$) Write $\Phi = (\phi_k)_k$ and $\Phi' = (\phi'_j)_j$. A matrix A with $A\Phi = \Phi'$ on P exists as soon as every linear relation between values of Φ also holds for Φ', i.e., $\sum_i \alpha_i \Phi(x_i) = 0 \Rightarrow \sum_i \alpha_i \Phi'(x_i) = 0$ for all finite families of points. Such a relation says that the two measures $\sigma^{\pm} = \sum_{i:\pm\alpha_i>0} |\alpha_i| \delta_{x_i}$ have the same $\mathcal{M}$-value. Then σ^- is compatible with $\mathcal{M}(\sigma^+)$, hence, since $\mathcal{M} \succeq \mathcal{M}'$, with $\mathcal{M}'(\sigma^+)$: $\mathcal{M}'(\sigma^+) = \mathcal{M}'(\sigma^-)$, which is the relation for Φ'. (iii) Immediate from the definitions. □

The characterization (ii) is pointwise: functions that agree almost everywhere with respect to area are not equivalent, since point loads see their values at single points. The number r of measurement functions is not monotone with respect to $\succeq$, because measurement functions can be redundant. The operator without measurement functions ($r = 0$, $\Pi = \mathbb{M}_+(P)$) is below every operator; its bounds are 0 and $+\infty$, and it violates Assumption 3.9.

Theorem 3.15 (Monotonicity with respect to information)**.** *Let $\mathcal{M} \succeq \mathcal{M}'$, $\nu_N \in \mathbb{M}_+(P)$, $y = \mathcal{M}(\nu_N)$ and $y' = \mathcal{M}'(\nu_N)$. Then, for every $q \in \mathbb{R}^2 \times \mathbb{R}$,*

$$\underline{h}(q; y', \mathcal{M}') \le \underline{h}(q; y, \mathcal{M}) \le h_{\mathcal{X}(\nu_N)}(q) \le \overline{h}(q; y, \mathcal{M}) \le \overline{h}(q; y', \mathcal{M}'), \tag{27}$$

and $\mathcal{X}^-(y', \mathcal{M}') \subseteq \mathcal{X}^-(y, \mathcal{M}) \subseteq \mathcal{X}(\nu_N) \subseteq \mathcal{X}^+(y, \mathcal{M}) \subseteq \mathcal{X}^+(y', \mathcal{M}')$. This applies in particular to the true measure $\nu_N^\star$.

Proof. $\nu_N \in \Pi_{\mathcal{M}}(y) \subseteq \Pi_{\mathcal{M}'}(y')$. An infimum (resp. supremum, intersection, union) over a smaller family is larger (resp. smaller, larger, smaller). □

Table 2 summarizes the measurement operators studied in the remainder of the report and the structure of the resulting problems, and Figure 1 illustrates what each of them observes.

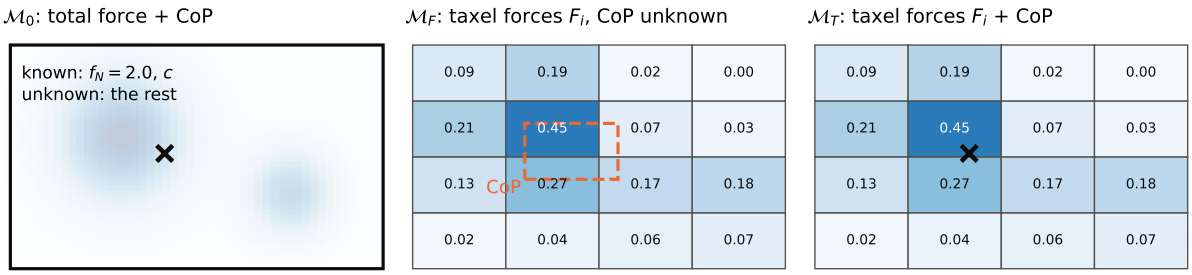


Figure 1: What each measurement operator observes for the same contact (4×4 taxels; notation of Section 5). $\mathcal{M}_0$: only f_N and the CoP, the pressure distribution (faded) is unknown. $\mathcal{M}_F$: the taxel forces F_i (cells filled with the mean pressure $F_i/|P_i|$); the CoP is only known to lie in the dashed rectangle $\bar{c}_F + R_0$, where $\bar{c}_F = \frac{1}{f_N}\sum_i F_i m_i$ is the force-weighted average of the taxel centers and $R_0 = [-\frac{w_x}{2}, \frac{w_x}{2}] \times [-\frac{w_y}{2}, \frac{w_y}{2}]$. Indeed, the local first moments lie in $F_i\overline{P_i}$, so the CoP lies in $\sum_i \frac{F_i}{f_N}\overline{P_i} = \bar{c}_F + R_0$, and point loads in the taxels reach every point of it. It has the diameter D of one taxel, whatever the forces, and an observed CoP outside it would contradict the taxel forces; the bounds do not use this set. $\mathcal{M}_T$: taxel forces and CoP.

4 $\mathcal{M}_0$ case (coarse measurement)

We now assume that the available measurements consist only of the resultant normal force and first moment. **Known:** P, μ, f_N and m_1. **Unknown:** how $\nu_N^\star$ is distributed over P. In this section P is convex.

Definition 4.1 (Coarse normal force measurement)**.** *Given the true (unknown) normal force measure $\nu_N^\star \in \mathbb{M}_+(P)$, the coarse measurement is its zeroth and first moments*

$$y_0 = \mathcal{M}_0(\nu_N^\star) \triangleq \left(\int_P d\nu_N^\star(x),\ \int_P x\, d\nu_N^\star(x) \right) = (f_N, m_1). \tag{28}$$

Its measurement functions are $\phi_1(x) = 1$, $\phi_2(x) = x_1$, $\phi_3(x) = x_2$, and it satisfies Assumption 3.9 with $\beta = (1, 0, 0)$.

Theorem 4.2 (Exact coarse lower bound)**.** *Given the coarse observation $y_0 = (f_N, m_1)$, for any $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,*

$$\boxed{\underline{h}(q; y_0, \mathcal{M}_0) = \mu \, \|f_N a + b J m_1\|_2} \tag{29}$$

For $f_N > 0$ this is $\mu f_N \, g(c; q)$: maximal dissipated power of normal force concentrated at the CoP.

Proof. By Lemma 3.12(iii), $h_{\mathcal{X}(\nu_N)}(q) \geq \mu \|f_N a + b J m_1\|_2$ for every $\nu_N \in \Pi_{\mathcal{M}_0}(y_0)$. If $f_N = 0$, then $\Pi_{\mathcal{M}_0}(y_0) = \{0\}$ and both sides vanish. Otherwise $c \in P$, so $f_N \delta_c \in \Pi_{\mathcal{M}_0}(y_0)$, and it attains the value. □

Corollary 4.3 (Exact guaranteed set; no torsion about the CoP)**.** *Let $f_N > 0$ and $c = m_1/f_N$. Then*

$$\mathcal{X}^-(y_0, \mathcal{M}_0) = \mathcal{X}(f_N \delta_c) = \{(f, c \times f) : \ \|f\|_2 \leq \mu f_N\}. \tag{30}$$

In particular, in CoP coordinates (Lemma 3.4), $T_c\mathcal{X}^-(y_0, \mathcal{M}_0) = f_N \, B(0, \mu) \times \{0\}$, with $B(0, \mu)$ the closed disk of radius μ: coarse measurement guarantees the friction for the tangential force, but no torsional moment about the CoP.

Proof. We show both inclusions. ($\subseteq$) The point load $f_N\delta_c$ is compatible, so $\mathcal{X}^- \subseteq \mathcal{X}(f_N\delta_c)$. With all the friction at c, its wrenches are $(f, c\times f)$ with $\|f\|_2 \le \mu f_N$. ($\supseteq$) Let $\|f\|_2 \le \mu f_N$ and let ν_N be any compatible measure. Mobilizing the same friction ratio f/f_N, of norm at most μ, at every contact point is admissible. Its resultant force is f, and its moment is $\int x\, d\nu_N \times f/f_N = m_1 \times f/f_N = c\times f$. So $(f, c\times f) \in \mathcal{X}(\nu_N)$ (compare Remark 4.2 of Yann's draft). □

Theorem 4.4 (Coarse upper bound)**.** *The supremum $\overline{h}(q; y_0, \mathcal{M}_0)$ is attained by a measure $\nu_N^+ \in \Pi_{\mathcal{M}_0}(y_0)$ supported on at most three points, i.e.,*

$$\nu_N^+ = \sum_{j=1}^{n} \lambda_j \delta_{x_j}, \qquad n \le 3, \qquad x_j \in P, \quad \lambda_j \ge 0, \qquad \sum_{j=1}^{n} \lambda_j = f_N, \qquad \sum_{j=1}^{n} \lambda_j x_j = m_1. \tag{31}$$

Proof. This is Proposition 4.4 of Yann's draft with $C = B(0,\mu)$ (Remark 4.6). It is an instance of the classical fact that extreme points of a set of measures defined by three linear moment constraints have at most three atoms [3, 2]. A direct argument has three steps (for $f_N = 0$ there is nothing to prove). *1. Geometric reformulation.* Let $\pi = \nu_N/f_N$ and $\Phi(x) = (x, g(x;q)) \in \mathbb{R}^3$. The constraints say that the first two components of $\int \Phi\, d\pi$ equal the CoP c, and the third is the objective divided by μf_N. Such averages fill the compact convex set $\mathcal{K} = \operatorname{conv}\Phi(P)$, so the bound is $\mu f_N t^\star$, where $(c, t^\star)$ is the highest point of $\mathcal{K}$ above c. *2. The maximizer lies on a face.* The point $(c, t^\star)$ is on the boundary of $\mathcal{K}$, so it lies on a supporting plane H, and $\mathcal{K}\cap H = \operatorname{conv}(\Phi(P)\cap H)$. *3. Carathéodory.* $\mathcal{K}\cap H$ has dimension at most two, so $(c, t^\star)$ is a convex combination $\sum_j \beta_j \Phi(x_j)$ of at most three points of $\Phi(P)$. The measure $f_N \sum_j \beta_j \delta_{x_j}$ is compatible and attains the bound. □

Corollary 4.5 (Polygonal contact patch)**.** *Let P be a convex polygon with vertices $v_1, \dots, v_n$. Then*

$$\boxed{\overline{h}(q; y_0, \mathcal{M}_0) = \mu \max_{\lambda \in \mathbb{R}^n_+} \left\{ \sum_{k=1}^{n} \lambda_k \|a + bJv_k\|_2 : \sum_{k=1}^{n} \lambda_k = f_N,\ \sum_{k=1}^{n} \lambda_k v_k = m_1 \right\}} \tag{32}$$

This is a linear program with three equality constraints, so it has an optimal solution with at most three nonzero weights.

Proof. Moving mass to the vertices. By Theorem 4.4, the supremum is attained by a measure with at most three atoms, so it suffices to consider such measures. Split the mass λ at an atom $x = \sum_k \alpha_k v_k$ (a convex combination of the vertices) into the masses $\lambda\alpha_k$ at the vertices v_k. This keeps f_N and m_1, and by convexity of $g(\cdot; q)$ (Lemma 3.12(i)) it does not decrease the objective. So the supremum is at most the value of the linear program. *Conversely,* a feasible λ gives the compatible measure $\sum_k \lambda_k \delta_{v_k}$, whose objective is the value of the linear program. *Three weights.* A linear program with three equality constraints has a basic optimal solution with at most three nonzero entries. □

Remark 4.6 (Connection with Yann's formulation)**.** *Theorem 4.4 and Corollary 4.5 are the Coulomb case $C = B(0,\mu)$ of Proposition 4.4 in Yann's draft, with the same reference point. With his notation $x^\perp = -Jx$, his moment $m_N = -\int\langle \rho_T, x^\perp\rangle$ is our τ_N, his twist (ω_N, v_T) is our $q = (v_T, \omega_N)$, and his integrand is $s_C(\omega_N x^\perp - v_T) = \mu\|v_T + \omega_N Jx\|_2 = \mu\, g(x;q)$. His set $f_N\, D(c^\perp)$ is $\mathcal{X}^+(y_0, \mathcal{M}_0)$ (components reordered), so that $f_N\, s_{D(c^\perp)}(-[\omega_N, v_T]) = \overline{h}(q; y_0, \mathcal{M}_0)$, and the concavity of $a \mapsto s_{D(a)}$ is Proposition 3.10(iii). His special cases are the CoP at a vertex of P (a single point load, both bounds equal $\mu f_N g(c;q)$) and $\omega_N = 0$ (both bounds equal $\mu f_N \|v_T\|_2$). As he notes, $D(\cdot)$ is the place to add priors on the distribution; the taxel measurements of the next sections are such priors.*

Example 4.7 (Segment patch). *Let* $P = [0, L] \times \{0\}$, $c = \left(\frac{L}{2}, 0\right)$, $\nu_N^{(1)} = f_N \delta_c$ *and* $\nu_N^{(2)} = \frac{f_N}{2}\delta_{(0,0)} + \frac{f_N}{2}\delta_{(L,0)}$. *Both have* $\mathcal{M}_0(\nu_N^{(1)}) = \mathcal{M}_0(\nu_N^{(2)}) = \left(f_N, f_N\left(\frac{L}{2}, 0\right)\right)$. *For the unit rotation about the CoP,* $q_c = (-Jc, 1)$, *so that* $g(x; q_c) = \|x - c\|_2$,

$$h_{\mathcal{X}(\nu_N^{(1)})}(q_c) = 0 \quad \neq \quad h_{\mathcal{X}(\nu_N^{(2)})}(q_c) = \mu f_N \frac{L}{2} = \overline{h}(q_c; y_0, \mathcal{M}_0). \tag{33}$$

The results of this section are recovered in Section 7 as the special case of a single taxel $P_1 = P$.

5 Taxel grids: setting and reduction lemma

We now assume that a tactile array measures the normal force carried by each taxel, with or without the coarse measurement. Tactile arrays are grids of rectangular taxels, and we restrict to them. Other cell shapes need barycentric weights in Lemma 5.5 instead of bilinear ones, and are not detailed. This section fixes the setting and collects the tools used in Sections 6 and 7.

Definition 5.1 (Taxel grid). *From now on, the patch is the sensor area* $P = [0, L] \times [0, \ell]$. *For* $k_x, k_y \geq 1$, *the grid* $G(k_x, k_y)$ *divides* P *into* $r = k_x k_y$ *taxels* $P_1, \ldots, P_r$: *the rectangles* $[\mathrm{lo}_i, \mathrm{hi}_i)$ *(componentwise half-open) of size* $w_x \times w_y = \frac{L}{k_x} \times \frac{\ell}{k_y}$, *those of the last column and row being closed on their outer side, so that every point of* P *belongs to exactly one taxel. We write* $\overline{P_i} = [\mathrm{lo}_i, \mathrm{hi}_i]$ *for the closed taxel,* m_i *for its center,* $v_{i1}, \ldots, v_{i4}$ *for its corners, and* $D = \sqrt{w_x^2 + w_y^2}$ *for the taxel diameter.*

Definition 5.2 (Taxel force measurement). *Given the true unknown normal force measure* $\nu_N^\star$ *and a taxel grid, the taxel force measurement is*

$$y_F = \mathcal{M}_F(\nu_N^\star) \triangleq (\nu_N^\star(P_1), \ldots, \nu_N^\star(P_r)), \tag{34}$$

with measurement functions $\phi_i = \mathbf{1}_{P_i}$, $i = 1, \ldots, r$. *The measured taxel forces are denoted* $F_i \triangleq \nu_N^\star(P_i)$, *and* $f_N = \sum_i F_i$. *Since* $\sum_i \mathbf{1}_{P_i} = 1$, $\mathcal{M}_F$ *satisfies Assumption 3.9. For the grid* $G(k_x, k_y)$ *we write* $\mathcal{M}_F(k_x, k_y)$ *(e.g. Yann's suggested grid is* $\mathcal{M}_F(2^{p_x}, 2^{p_y})$*).*

Definition 5.3 (Combined coarse and taxel force measurement). *The combined measurement is the concatenation* $\mathcal{M}_T \triangleq \mathcal{M}_0 \vee \mathcal{M}_F$,

$$y_T = \mathcal{M}_T(\nu_N^\star) = (f_N, m_1, F_1, \ldots, F_r), \qquad \Pi_{\mathcal{M}_T}(y_T) = \Pi_{\mathcal{M}_0}(y_0) \cap \Pi_{\mathcal{M}_F}(y_F). \tag{35}$$

We write $\mathcal{M}_T(k_x, k_y) \triangleq \mathcal{M}_0 \vee \mathcal{M}_F(k_x, k_y)$

Proposition 5.4 (Information chains). *(i)* $\mathcal{M}_F(k_x, k_y) \succeq \mathcal{M}_F(k'_x, k'_y)$ *if and only if* k'_x *divides* k_x *and* k'_y *divides* k_y. *The same holds for* $\mathcal{M}_T$. *In particular, dyadic grids form chains, while* 3×3 *and* 4×4 *grids are not comparable. Moreover* $\mathcal{M}_F(1, 1)$ *measures the resultant force* f_N *only, and* $\mathcal{M}_T(1, 1)$ *and* $\mathcal{M}_0$ *are equivalent.*

(ii) $\mathcal{M}_T \succeq \mathcal{M}_0$ *and* $\mathcal{M}_T \succeq \mathcal{M}_F$. *If* $r \geq 2$, $\mathcal{M}_0$ *and* $\mathcal{M}_F$ *are not comparable.*

Proof. Recall that $\mathcal{M} \succeq \mathcal{M}'$ if and only if every measurement function of $\mathcal{M}'$ is a linear combination of those of $\mathcal{M}$ (Proposition 3.14). (i) If $k'_x \mid k_x$ and $k'_y \mid k_y$, each coarse taxel is a union of fine taxels, so its indicator is the sum of the indicators of the fine taxels it contains. Conversely, suppose for example that $k'_x \nmid k_x$. Then some edge between two coarse taxels lies strictly inside a fine taxel. The indicator of a coarse taxel jumps across this edge, whereas the fine indicators are constant there and $1, x_1, x_2$ are continuous, so it is not a linear combination of them. The case $k'_y \nmid k_y$ is the same. (ii) $\mathcal{M}_T$ has the measurement functions of both. If $r \geq 2$, a taxel indicator is not affine, so $\mathcal{M}_0 \not\succeq \mathcal{M}_F$, and x_1 is not constant on a taxel, so $\mathcal{M}_F \not\succeq \mathcal{M}_0$. □

Lemma 5.5 (Concentration and spreading). *Let $\nu_N \in \mathbb{M}_+(P)$ have taxel forces $F_i = \nu_N(P_i)$ and local first moments $u_i \triangleq \int_{P_i} x\, d\nu_N(x)$, so that $\sum_i F_i = f_N$ and $\sum_i u_i = m_1$. There are weights $\lambda_{ik} \geq 0$ with $\sum_k \lambda_{ik} = F_i$ and $\sum_k \lambda_{ik} v_{ik} = u_i$ (and $\lambda_{ik} = 0$ if $F_i = 0$) such that, for every twist $q = (a, b)$,*

$$\underbrace{\mu \sum_{i:F_i>0} F_i\, g\Big(\tfrac{u_i}{F_i}; q\Big)}_{h_{\mathcal{X}(\nu_N^c)}(q)} \;\leq\; h_{\mathcal{X}(\nu_N)}(q) \;\leq\; \underbrace{\mu \sum_{i,k} \lambda_{ik}\, g(v_{ik}; q)}_{h_{\mathcal{X}(\nu_N^s)}(q)}. \tag{36}$$

The outer terms are the values of $h_{\mathcal{X}(\cdot)}(q)$ for the concentrated *load $\nu_N^c = \sum_i F_i \delta_{u_i/F_i}$ and the* spread *load $\nu_N^s = \sum_{i,k} \lambda_{ik} \delta_{v_{ik}}$. Both put the force F_i in the closed taxel $\overline{P_i}$ and have the first moment m_1 of ν_N.*

Proof. By Proposition 3.2,

$$h_{\mathcal{X}(\nu_N)}(q) = \mu \sum_i \int_{P_i} g(x; q)\, d\nu_N(x),$$

and $g(\cdot; q)$ is convex (Lemma 3.12(i)). Fix a taxel with $F_i > 0$. *Concentration.* $\nu_N|_{P_i}/F_i$ is a probability measure with mean u_i/F_i, so Jensen's inequality gives $\int_{P_i} g\, d\nu_N \geq F_i\, g(u_i/F_i; q)$. *Spreading.* For $x \in \overline{P_i}$, let $\xi = (x_1 - \mathrm{lo}_{i,1})/w_x$ and $\eta = (x_2 - \mathrm{lo}_{i,2})/w_y$, and let $\alpha_{ik}(x)$ be the bilinear weights of the four corners, $(1-\xi)(1-\eta)$, $\xi(1-\eta)$, $\xi\eta$, $(1-\xi)\eta$ (counterclockwise from lo_i). They are nonnegative, sum to one, and satisfy $\sum_k \alpha_{ik}(x) v_{ik} = x$. By convexity, $g(x; q) \leq \sum_k \alpha_{ik}(x)\, g(v_{ik}; q)$. Integrating against $\nu_N|_{P_i}$ gives the right inequality with $\lambda_{ik} = \int_{P_i} \alpha_{ik}\, d\nu_N$. Integrating $\sum_k \alpha_{ik} = 1$ and $\sum_k \alpha_{ik} v_{ik} = x$ in the same way gives $\sum_k \lambda_{ik} = F_i$ and $\sum_k \lambda_{ik} v_{ik} = u_i$. Summing over the taxels proves the claim. □

Interpretation. The lemma replaces the unknown measure, taxel by taxel, by finitely many point loads that keep the data of the measurement (F_i and m_1), and brackets its friction power in every direction (Figure 2).

- *Concentrate* (left inequality): placing the force of each taxel at its local CoP u_i/F_i can only decrease the friction power. This gives the lower bounds of $\mathcal{M}_F$ (Theorem 6.1) and $\mathcal{M}_T$ (Theorem 7.1); for $\mathcal{M}_0$ it is the single point load at the CoP (Theorem 4.2).
- *Spread* (right inequality): splitting the force of each taxel among its corners, with the weights of the lemma, can only increase it. This gives the upper bounds of $\mathcal{M}_F$ and $\mathcal{M}_T$ (Theorems 6.1 and 7.2); for $\mathcal{M}_0$ it is the reduction to the vertices of P (Corollary 4.5), i.e., Yann's reduction to the extreme points (his Eq. (33) and Proposition 4.4), applied taxel by taxel.

6 $\mathcal{M}_F$ case (taxel forces)

Known: the taxel forces F_i. **Unknown:** the distribution of $\nu_N^\star$ inside each taxel, hence also its CoP. Without the CoP constraint, the taxels decouple and both bounds have closed forms.

Theorem 6.1 (Exact bounds for taxel force measurements). *On a taxel grid, for every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,*

$$\boxed{\underline{h}(q; y_F, \mathcal{M}_F) = \mu \sum_{i=1}^{r} F_i \inf_{x \in P_i} \|a + bJx\|_2, \qquad \overline{h}(q; y_F, \mathcal{M}_F) = \mu \sum_{i=1}^{r} F_i \sup_{x \in P_i} \|a + bJx\|_2.} \tag{37}$$

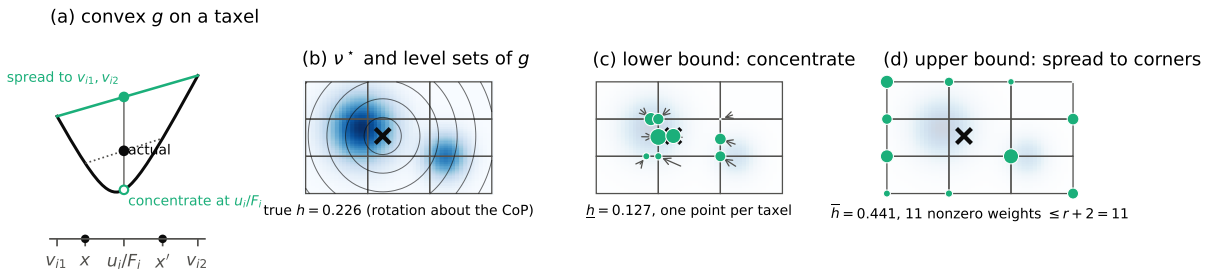


Figure 2: Concentrate vs. spread (Lemma 5.5, Eq. (36)), 3×3 taxels, unit rotation about the CoP $q_c = (-Jc, 1)$, so $g(x; q_c) = \|x - c\|_2$. (a) For two masses at x, x' of a taxel, with local CoP u_i/F_i: concentrating at u_i/F_i lowers the objective (Jensen), spreading to the corners v_{i1}, v_{i2} raises it (chord). (b) True measure and level sets of g. (c) Minimizer of Theorem 7.1: one Dirac per taxel, pulled from the local CoPs of the taxels toward the CoP (arrows). (d) Maximizer of Theorem 7.2: weights on the taxel corners, pushed away from the CoP, with at most $r+2$ nonzero weights. Marker areas are proportional to force.

The supremum equals $\max_{k=1,\dots,4} \|a + bJv_{ik}\|_2$*, the maximum over the corners of taxel* i*. For* $b \neq 0$*, with the instantaneous center of rotation* $p = Ja/b$*, the bounds are* $\mu|b| \sum_i F_i\, d_i^{\min}$ *and* $\mu|b| \sum_i F_i\, d_i^{\max}$*, where* $d_i^{\min}$ *and* $d_i^{\max}$ *are the smallest and largest distances from* p *to the taxel. With* $\Delta_i = m_i - p$*,*

$$d_i^{\min} = \left\| \left((|\Delta_{i,1}| - \tfrac{w_x}{2})_+ , (|\Delta_{i,2}| - \tfrac{w_y}{2})_+ \right) \right\|_2 , \qquad d_i^{\max} = \left\| \left(|\Delta_{i,1}| + \tfrac{w_x}{2}, |\Delta_{i,2}| + \tfrac{w_y}{2} \right) \right\|_2 . \tag{38}$$

For $b = 0$ *both bounds equal* $\mu f_N \|a\|_2$*.*

Proof. Since only the taxel forces are measured, a measure is compatible if and only if its part in each taxel P_i has mass F_i, and these parts are free. The objective $\mu \sum_i \int_{P_i} g(\cdot; q)\, d\nu_N$ is a sum of one term per taxel, so each term is optimized on its own. For one taxel, $F_i \inf_{P_i} g \leq \int_{P_i} g\, d\nu_N \leq F_i \sup_{P_i} g$, and point loads at points of P_i approach both ends. This gives (37). The function $g(\cdot; q)$ is continuous and convex, so its infimum over P_i is its minimum over the closed rectangle, and its supremum is the maximum over the rectangle, attained at a corner. For $b \neq 0$, $g(x; q) = |b|\, \|x - p\|_2$: the closed forms are the distance from p to the rectangle, and to its farthest corner. □

7 $\mathcal{M}_T$ case (taxel forces and CoP)

Known: the taxel forces F_i and the first moment m_1, i.e., the CoP. **Unknown:** the distribution of $\nu_N^\star$ inside each taxel. The taxel constraints act on each taxel separately, but the first-moment constraint $\int_P x\, d\nu_N = m_1$ couples the taxels.

Theorem 7.1 (Exact lower bound for combined measurements)**.** *On a taxel grid, given* $y_T = (f_N, m_1, F_1, \dots, F_r) = \mathcal{M}_T(\nu_N^\star)$*, for every* $q = (a, b)$*,*

$$\begin{aligned} \underline{h}(q; y_T, \mathcal{M}_T) = \mu \min_{u_1,\dots,u_r \in \mathbb{R}^2} \quad & \sum_{i=1}^{r} \|F_i a + bJu_i\|_2 \\ \textit{subject to} \quad & F_i \,\mathrm{lo}_i \leq u_i \leq F_i \,\mathrm{hi}_i, \qquad i = 1, \dots, r, \\ & \sum_{i=1}^{r} u_i = m_1. \end{aligned} \tag{39}$$

The variables are the local first moments (inequalities componentwise): for $F_i > 0$, $u_i = F_i c_i$ where c_i is the local CoP of taxel i, constrained to lie in the taxel, and the objective is $\mu \sum_i F_i\, g(c_i; q)$, while $F_i = 0$ forces $u_i = 0$. The problem is an SOCP.

Proof. ($\geq$) For a compatible ν_N, the local first moments $u_i = \int_{P_i} x\, d\nu_N$ are feasible, and $\int_{P_i} g\, d\nu_N \geq \|F_i a + bJu_i\|_2$ by Lemma 3.12(iii) applied to the restriction of ν_N to P_i. So $h_{\mathcal{X}(\nu_N)}(q)$ is at least the minimum. ($\leq$) Let u be optimal. The point loads $\sum_{i:F_i>0} F_i \delta_{u_i/F_i}$ have the taxel forces F_i, the first moment m_1, and the optimal value. But a point u_i/F_i may lie on an upper edge of its taxel, which the half-open P_i does not contain. We move every point of taxel i slightly toward the local CoP $c_i^\star$ of $\nu_N^\star$ in P_i, by $x \mapsto (1-\varepsilon)x + \varepsilon c_i^\star$. The new point lies in P_i, since $c_i^\star \in P_i$. The taxel forces are unchanged, and so is the first moment, because $\sum_i F_i c_i^\star = m_1$. So the new measure is compatible, and by Lemma 3.12(ii) its value is within $\varepsilon\mu|b|D\sum_i F_i$ of the optimum. Let $\varepsilon \to 0$. The feasible set is compact and nonempty (it contains the local first moments of $\nu_N^\star$), and the objective is convex and continuous; epigraph variables give an SOCP. □

Theorem 7.2 (Exact upper bound for combined measurements)**.** *Under the assumptions of Theorem 7.1, for every $q = (a, b)$,*

$$\begin{aligned} \overline{h}(q; y_T, \mathcal{M}_T) = \mu \max_{\lambda_{ik} \geq 0} \quad & \sum_{i=1}^{r} \sum_{k=1}^{4} \lambda_{ik} \, \|a + bJv_{ik}\|_2 \\ \text{subject to} \quad & \sum_{k=1}^{4} \lambda_{ik} = F_i, \qquad i = 1, \ldots, r, \\ & \sum_{i=1}^{r} \sum_{k=1}^{4} \lambda_{ik} v_{ik} = m_1. \end{aligned} \tag{40}$$

This is a linear program, and an optimal solution exists with at most $r + 2$ nonzero weights.

Proof. ($\leq$) For a compatible ν_N, Lemma 5.5 gives corner weights $\lambda_{ik} \geq 0$ with $\sum_k \lambda_{ik} = F_i$ and $\sum_{i,k} \lambda_{ik} v_{ik} = \sum_i u_i = m_1$. They are feasible, and $h_{\mathcal{X}(\nu_N)}(q) \leq \mu \sum_{i,k} \lambda_{ik} g(v_{ik}; q)$, the objective of λ. ($\geq$) Let λ be feasible. The corner loads $\sum_{i,k} \lambda_{ik} \delta_{v_{ik}}$ have the taxel forces F_i, the first moment m_1, and the objective value, but a corner may lie on an edge owned by a neighboring taxel. As in the proof of Theorem 7.1, moving every point of taxel i by $x \mapsto (1-\varepsilon)x + \varepsilon c_i^\star$, with $c_i^\star$ the local CoP of $\nu_N^\star$ in P_i, gives a compatible measure whose value is within $\varepsilon\mu|b|D\sum_i F_i$ of the objective. Let $\varepsilon \to 0$. The LP is feasible (the corner weights of $\nu_N^\star$ are) and bounded, and a basic optimal solution has at most as many nonzero entries as equality constraints, i.e., $r + 2$. □

Remark 7.3 (Recovering the coarse and taxel-force cases)**.** *For a single taxel ($k_x = k_y = 1$), the constraint of (39) forces $u_1 = m_1$, and Theorem 7.1 reduces to Theorem 4.2; Theorem 7.2 reduces to Corollary 4.5 for the rectangle P, with $r + 2 = 3$ nonzero weights. Dropping the first-moment constraint in (39) and (40) decouples the taxels and gives the $\mathcal{M}_F$ bounds of Theorem 6.1. Consistently with Theorem 3.15 and $\mathcal{M}_T \succeq \mathcal{M}_F$, the $\mathcal{M}_T$ bounds lie between the $\mathcal{M}_F$ bounds:*

$$\underline{h}(q; y_F, \mathcal{M}_F) \leq \underline{h}(q; y_T, \mathcal{M}_T) \leq \overline{h}(q; y_T, \mathcal{M}_T) \leq \overline{h}(q; y_F, \mathcal{M}_F). \tag{41}$$

The difference is the value of measuring the CoP in addition to the taxel forces.

8 Comparison of measurements and tightening under refinement

All results of this section follow from Theorem 3.15, the information chains of Proposition 5.4, and the closed forms of Theorem 6.1.

Proposition 8.1 (Comparison of coarse, taxel and combined measurements)**.** *For every twist* $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,

$$\boxed{\begin{aligned}\max\{\underline{h}(q; y_0, \mathcal{M}_0),\, \underline{h}(q; y_F, \mathcal{M}_F)\} \le \underline{h}(q; y_T, \mathcal{M}_T) \le h_{\mathcal{X}(\nu_N^\star)}(q),\\ h_{\mathcal{X}(\nu_N^\star)}(q) \le \overline{h}(q; y_T, \mathcal{M}_T) \le \min\left\{\overline{h}(q; y_0, \mathcal{M}_0),\, \overline{h}(q; y_F, \mathcal{M}_F)\right\},\end{aligned}} \tag{42}$$

with $\underline{h}(q; y_0, \mathcal{M}_0) = \mu\|f_N a + bJm_1\|_2$*. The bounds of* $\mathcal{M}_0$ *and* $\mathcal{M}_F$ *are not ordered in general, in either direction.*

Proof. Since $\mathcal{M}_T = \mathcal{M}_0 \vee \mathcal{M}_F$, $\mathcal{M}_T$ is at least as informative as both (Proposition 3.14(iii)), and Theorem 3.15 gives the chain. The bounds of $\mathcal{M}_0$ and $\mathcal{M}_F$ are not ordered, in either direction. With a single taxel, $\mathcal{M}_F$ measures only f_N, so $\mathcal{M}_0$ is tighter, e.g. $\underline{h}(q; y_F, \mathcal{M}_F) = \mu f_N|b|\operatorname{dist}(p, P) < \underline{h}(q; y_0, \mathcal{M}_0) = \mu f_N|b|\|c - p\|_2$ when the CoP c is not the point of P closest to p. With fine taxels, $\mathcal{M}_F$ is tighter: its bounds converge to the truth (Proposition 8.3), while those of $\mathcal{M}_0$ do not in general (Example 4.7). □

Proposition 8.2 (Tightening under taxel refinement)**.** *Let* $\mathcal{M}'_F$, $\mathcal{M}'_T$ *and* $\mathcal{M}_F$, $\mathcal{M}_T$ *be the taxel-force and combined measurements of the grids* $G(k'_x, k'_y)$ *and* $G(k_x, k_y)$*, with* $k_x \mid k'_x$ *and* $k_y \mid k'_y$ *(e.g. dyadic grids, each doubling the previous one). Then, for every* q *and* $\mathcal{M} \in \{\mathcal{M}_F, \mathcal{M}_T\}$*, with* $y = \mathcal{M}(\nu_N^\star)$ *and* $y' = \mathcal{M}'(\nu_N^\star)$*,*

$$\boxed{\underline{h}(q; y, \mathcal{M}) \le \underline{h}(q; y', \mathcal{M}') \le h_{\mathcal{X}(\nu_N^\star)}(q) \le \overline{h}(q; y', \mathcal{M}') \le \overline{h}(q; y, \mathcal{M}),} \tag{43}$$

and the guaranteed sets grow and the possible sets shrink.

Proof. Proposition 5.4(i) and Theorem 3.15. □

The monotonicity is associated with **nested grids**, not with the number of taxels r: a 3×3 and a 4×4 grid carry different information (Proposition 5.4(i)), and nothing guarantees that the second gives tighter bounds. The gap bounds below, on the other hand, only require small taxel diameters.

Proposition 8.3 (Gap bound)**.** *For every twist* $q = (a, b)$ *and* $\mathcal{M} \in \{\mathcal{M}_F, \mathcal{M}_T\}$ *on a grid of taxel diameter* D*,*

$$\boxed{0 \le \overline{h}(q; y, \mathcal{M}) - \underline{h}(q; y, \mathcal{M}) \le \mu|b|\, f_N\, D.} \tag{44}$$

Hence, along any sequence of grids with $D \to 0$*, the four bounds of* $\mathcal{M}_F$ *and* $\mathcal{M}_T$ *converge to* $h_{\mathcal{X}(\nu_N^\star)}(q)$*.*

Proof. By Theorem 6.1, the $\mathcal{M}_F$ gap is $\mu \sum_i F_i \left(\sup_{P_i} g - \inf_{P_i} g\right)$. The function $g(\cdot; q)$ is $|b|$-Lipschitz (Lemma 3.12(ii)), so each bracket is at most $|b|D$ and the sum is at most $\mu|b|f_N D$. The $\mathcal{M}_T$ bounds lie inside the $\mathcal{M}_F$ bounds (41), so their gap is smaller. Convergence follows since $h_{\mathcal{X}(\nu_N^\star)}(q)$ lies between the bounds. □

The next result answers how much one refinement step tightens the bounds. For a twist $q = (a, b)$ with $b \neq 0$, the speed field $g(x) = \|a + bJx\|_2 = |b|\,\|x - p\|_2$ is $|b|$ times the distance to the point $p = Ja/b$ where the speed vanishes (the instantaneous center of rotation). It is continuously differentiable away from p, with $\|\nabla g\|_2 = |b|$.

Theorem 8.4 (First-order rate of tightening)**.** *Consider grids of square taxels ($w_x = w_y$), and let $\nu_N^\star$ have a density with respect to area. Then, for every twist $q = (a, b)$ with $b \neq 0$,*

$$\lim_{w_x \to 0} \frac{\overline{h}(q; y_F, \mathcal{M}_F) - \underline{h}(q; y_F, \mathcal{M}_F)}{w_x} = \mu \int_P (|\partial_1 g| + |\partial_2 g|)\, d\nu_N^\star = \mu |b|\, C_\nu, \qquad C_\nu \in \left[f_N, \sqrt{2} f_N \right]. \tag{45}$$

So, if $f_N > 0$, doubling the number of taxels per side divides the $\mathcal{M}_F$ gap by a factor that tends to 2, and the bound $\mu|b| f_N D$ of Proposition 8.3 ($D = \sqrt{2} w_x$) is asymptotically sharp up to a factor $\sqrt{2}$. Since the $\mathcal{M}_T$ gap is at most the $\mathcal{M}_F$ gap (41), the same limit bounds it from above.

Proof. Gap as an integral. By Theorem 6.1, the gap is $\mu \sum_i F_i\, \theta_i$, where θ_i is the range of g over the taxel P_i. Since $F_i = \nu_N^\star(P_i)$, this is $\mu \int_P \theta(x)\, d\nu_N^\star(x)$, where $\theta(x)$ is the range of g over the taxel that contains x. *Pointwise limit.* Fix $x \neq p$. Near x, g is continuously differentiable, so on the shrinking taxel that contains x it equals its linearization at x up to $o(w_x)$. The range of a linear function over a square of side w_x is $w_x(|\partial_1 g(x)| + |\partial_2 g(x)|)$, so $\theta(x)/w_x \to |\partial_1 g(x)| + |\partial_2 g(x)|$. *Domination.* g is $|b|$-Lipschitz, so $\theta(x)/w_x \leq |b|\sqrt{2}$. Since $\nu_N^\star$ has a density, the point p has measure zero, and dominated convergence gives the limit. *Constant.* $\nabla g/|b|$ is a unit vector, so $|b| \leq |\partial_1 g| + |\partial_2 g| \leq \sqrt{2}|b|$, which gives $C_\nu \in [f_N, \sqrt{2} f_N]$. □

9 Certified load factor

A controller does not need the bounds direction by direction. It needs to know whether a given load w can be carried, whether it certainly cannot, and by how much the contact should be strengthened (e.g., by squeezing harder). This is a question about the ray $\{sw : s \geq 0\}$: how far along it the guaranteed and the possible sets extend. There are two ways to use the support function $\overline{h}$ of the convex possible set $\mathcal{X}^+$ (Proposition 3.10) to test whether w belongs to it. *Projection:* the Frank–Wolfe method computes the distance from w to $\mathcal{X}^+$ using, at each iteration, only a support point of $\mathcal{X}^+$ (a measure attaining $\overline{h}(q)$), as the Gilbert–Johnson–Keerthi (GJK) algorithm does for collision shapes; a direction q with $\langle q, w \rangle > \overline{h}(q)$ proves that w is outside and stops the iteration. *Ray:* the *upper load factor* $\overline{s}$ is how far the ray extends in $\mathcal{X}^+$. It is a unitless ratio (the factor by which the contact must be strengthened) and equals the smallest value of $\overline{h}(q)$ over the directions with $\langle q, w \rangle = 1$. For the taxel measurements $\overline{h}$ is explicit and convex, so the ray view is one convex program, with no iteration. The guaranteed set $\mathcal{X}^-$ has no such oracle, since its support function is the convex envelope of $\underline{h}$, but the ray view still works: the *certified load factor* $\underline{s}$ is the smallest value of $\underline{h}(q)$ over $\langle q, w \rangle = 1$. This minimization is nonconvex (a finite set of directions would overestimate it), and a change of variables makes it convex (Proposition 9.4). Figure 3 illustrates both views.

Definition 9.1 (Load factors)**.** *Let $w = (f, \tau) \in \mathbb{R}^2 \times \mathbb{R}$ be a nonzero load (about the origin). The load factor (capacity) of a contact with normal force ν_N is $s^\star(\nu_N; w) \triangleq \max\{s \geq 0 :\ sw \in \mathcal{X}(\nu_N)\}$. Given a measurement $y = \mathcal{M}(\nu_N^\star)$, the certified (lower) and the upper load factors are how far the ray of w extends in the guaranteed and in the possible set,*

$$\underline{s}(w; y, \mathcal{M}) \triangleq \max\left\{ s \geq 0 :\ sw \in \mathcal{X}^-(y, \mathcal{M}) \right\}, \tag{46}$$

$$\overline{s}(w; y, \mathcal{M}) \triangleq \max\left\{ s \geq 0 :\ sw \in \operatorname{cl} \mathcal{X}^+(y, \mathcal{M}) \right\}. \tag{47}$$

Since $\mathcal{X}(t\nu_N) = t\, \mathcal{X}(\nu_N)$, a contact whose normal force is scaled by t has load factors scaled by t.

Proposition 9.2 (Load factors from the support functions)**.** *Let $w \neq 0$ and $\mathcal{M}$ satisfy Assumption 3.9.*

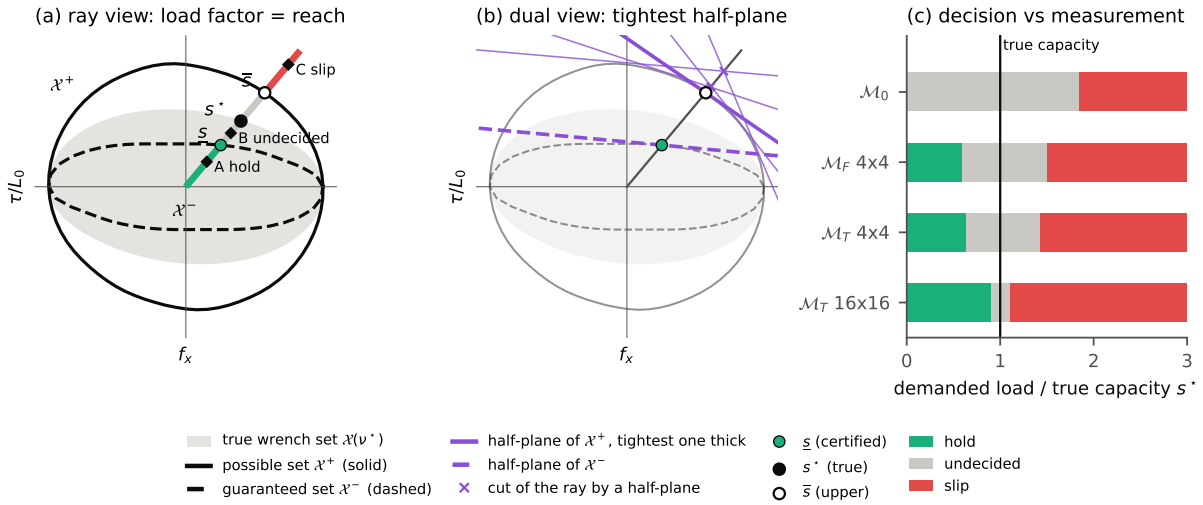


Figure 3: Load factors for one contact (4×4 taxels, $\mathcal{M}_T$) and a load $w = (f_x, 0, \tau)$ in the plane $f_y = 0$ of the wrench space (the torque is plotted as τ/L_0 with a lever $L_0 = 0.4$). Possible set $\mathcal{X}^+$: solid; guaranteed set $\mathcal{X}^-$: dashed; true set: grey. (a) Ray view: w is carried by every compatible measure up to $\underline{s}$ (green), undecided up to $\overline{s}$ (grey), and slips beyond (red); $s^\star$ is the true factor, and A, B, C are loads with the three verdicts. (b) Dual view (purple): each direction q with $\langle q, w\rangle = 1$ gives a half-plane containing $\mathcal{X}^+$ that cuts the ray at $\overline{h}(q)$ (crosses); $\overline{s}$ is the tightest cut (thick solid line, tangent to $\mathcal{X}^+$), and the thick dashed line is tangent to $\mathcal{X}^-$ at $\underline{s}\, w$. (c) The undecided band shrinks as the measurement becomes more informative.

(i) The load factors are the smallest values of the support bounds over the directions q with $\langle q, w\rangle = 1$:

$$\boxed{\underline{s}(w; y, \mathcal{M}) = \inf_{q:\, \langle q,w\rangle = 1} \underline{h}(q; y, \mathcal{M}), \qquad \overline{s}(w; y, \mathcal{M}) = \inf_{q:\, \langle q,w\rangle = 1} \overline{h}(q; y, \mathcal{M}).} \tag{48}$$

(ii) They bracket the load factor of every compatible contact, in particular of the true one: $\underline{s}(w; y, \mathcal{M}) = \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} s^\star(\nu_N; w) \le s^\star(\nu_N^\star; w) \le \sup_{\nu_N \in \Pi_{\mathcal{M}}(y)} s^\star(\nu_N; w) \le \overline{s}(w; y, \mathcal{M})$.

Proof. (i) By Proposition 3.10(ii), $sw \in \mathcal{X}^-(y)$ if and only if $\langle q, sw\rangle \le \underline{h}(q)$ for all q. Directions with $\langle q, w\rangle \le 0$ impose nothing, since $\underline{h} \ge 0$, and the others give $s \le \underline{h}(q)/\langle q, w\rangle$. By positive homogeneity, the smallest ratio is $\inf_{\langle q,w\rangle = 1} \underline{h}(q)$. The same holds for $\operatorname{cl} \mathcal{X}^+(y)$, which contains 0, is bounded by Assumption 3.9, and has support function $\overline{h}$. (ii) For a fixed measure, $\mathcal{X}(\nu_N)$ is convex, compact and contains 0, so $\{s \ge 0 : sw \in \mathcal{X}(\nu_N)\} = [0, s^\star(\nu_N; w)]$. Hence $sw \in \mathcal{X}^-(y)$ if and only if $s \le s^\star(\nu_N; w)$ for every compatible ν_N (the equality), and if $s \le s^\star(\nu_N; w)$ for one compatible ν_N, then $sw \in \mathcal{X}(\nu_N) \subseteq \mathcal{X}^+(y)$ (the last inequality). □

Corollary 9.3 (Decision rule)**.** *Let $w \ne 0$. If $\underline{s} \ge 1$, every compatible measure carries w (hold). If $\overline{s} < 1$, none does and the contact slips whatever the actual distribution (slip). Otherwise the measurement does not decide (undecided). At $\underline{s} = 1$ the load is carried, at the onset of sliding for the least favorable compatible measure, and $\underline{s} > 1$ gives a strict margin. If $\mathcal{M} \succeq \mathcal{M}'$, then* $\underline{s}(w; y', \mathcal{M}') \le \underline{s}(w; y, \mathcal{M}) \le s^\star(\nu_N^\star; w) \le \overline{s}(w; y, \mathcal{M}) \le \overline{s}(w; y', \mathcal{M}')$.

Proof. The three cases follow from Proposition 9.2(ii), and the ordering from the nesting of the sets in Theorem 3.15. □

The next two propositions give the convex programs for the taxel measurements.

Proposition 9.4 (Certified load factor for taxel measurements)**.** *On a taxel grid (Definition 5.1), let $\mathcal{M} \in \{\mathcal{M}_F, \mathcal{M}_T\}$. Then $\underline{s}(w; y, \mathcal{M}) = \min\{V_{+1}, V_{-1}\}$, where for $\sigma \in \{-1, +1\}$*

$$
\begin{aligned}
V_\sigma \triangleq \mu \inf_{a\in\mathbb{R}^2,\, b\in\mathbb{R},\, \zeta_i\in\mathbb{R}^2} \quad & \sum_{i=1}^{r} F_i \left\| a + J\zeta_i \right\|_2 \\
\text{subject to} \quad & a^\top f + b\tau = 1, \qquad \sigma b \geq 0, \\
& \sigma\left(\zeta_i - b\,\mathrm{lo}_i\right) \geq 0, \quad \sigma\left(b\,\mathrm{hi}_i - \zeta_i\right) \geq 0, \qquad i = 1, \ldots, r, \\
& \textstyle\sum_{i=1}^{r} F_i \zeta_i = b\, m_1 \qquad (\mathcal{M}_T \text{ only}),
\end{aligned}
\tag{49}
$$

with $V_\sigma = +\infty$ if infeasible. Each V_σ is an SOCP. The minimizer gives the local CoPs $c_i = \zeta_i / b$ of a worst-case point load per taxel $\hat{\nu}_N = \sum_i F_i \delta_{c_i}$ with $s^\star(\hat{\nu}_N; w) = \underline{s}$ (any admissible c_i if $b = 0$); it is compatible when each c_i lies in its own taxel.

Proof. Bilinear form. By Proposition 9.2(i), $\underline{s}$ is the infimum of $\underline{h}(q)$ over $\langle q, w\rangle = 1$, and for a fixed twist $\underline{h}(q) = \mu \sum_i F_i \|a + bJc_i\|_2$ minimized over the local CoPs c_i of the taxels (Theorems 6.1 and 7.1), i.e., $\mathrm{lo}_i \leq c_i \leq \mathrm{hi}_i$ and $\sum_i F_i c_i = m_1$ for $\mathcal{M}_T$. So $\underline{s}$ is a joint minimum over (q, c_i) of an objective that is bilinear in (b, c_i). *Substitution.* Set $\zeta_i = bc_i$ (a homogenization of the box constraint). Then $bJc_i = J\zeta_i$, so the objective is linear inside the norms, and multiplying the constraints by b gives the linear constraints of V_σ. Multiplying by b flips an inequality when $b < 0$, which is why the two signs of b are treated separately; for $\sigma b > 0$ the map $c_i \mapsto \zeta_i$ is a bijection. For $b = 0$, both objectives equal $\mu f_N \|a\|_2$ (the box constraints force $\zeta_i = 0$). For fixed σ the problem minimizes a sum of norms of affine functions under linear constraints, an SOCP. *Worst case.* If $\hat{\nu}_N$ is compatible, $\underline{s} \leq s^\star(\hat{\nu}_N; w)$. With the optimal $q^\star = (a^\star, b^\star)$, Proposition 9.2(i) gives $s^\star(\hat{\nu}_N; w) \leq h_{\mathcal{X}(\hat{\nu}_N)}(q^\star) = \mu \sum_i F_i \|a^\star + b^\star J c_i\|_2 = \underline{s}$. □

Proposition 9.5 (Upper load factor for taxel measurements)**.** *On a taxel grid, with the taxel corners v_{ik}, the upper load factor is the SOCP*

$$
\begin{aligned}
\overline{s}(w; y, \mathcal{M}) = \max_{s,\, \lambda_{ik}\geq 0,\, \kappa_{ik}} \quad s \quad \text{s.t.} \quad & \textstyle\sum_k \lambda_{ik} = F_i, \quad \|\kappa_{ik}\|_2 \leq \mu\lambda_{ik}, \\
& \textstyle\sum_{i,k} \kappa_{ik} = s\, f, \quad \sum_{i,k} v_{ik} \times \kappa_{ik} = s\, \tau, \\
& \textstyle\sum_{i,k} \lambda_{ik} v_{ik} = m_1 \quad (\mathcal{M}_T \text{ only}).
\end{aligned}
\tag{50}
$$

It is the largest s such that friction forces κ_{ik} at the corners of a feasible corner load λ balance $s\,w$. The optimal λ is a most favorable corner load.

Proof. A feasible corner load λ carries the wrenches $(\sum \kappa, \sum v \times \kappa)$ with $\|\kappa_{ik}\|_2 \leq \mu\lambda_{ik}$. Each such wrench belongs to $\mathrm{cl}\,\mathcal{X}^+(y)$, because a corner load is a limit of compatible measures (the boundary push in the proofs of Theorems 7.1 and 7.2). Conversely, for a compatible ν_N the spreading inequality of Lemma 5.5 gives a feasible corner load whose support function dominates that of ν_N, so its wrenches contain $\mathcal{X}(\nu_N)$. Hence $\mathrm{cl}\,\mathcal{X}^+(y)$ is the set of wrenches of the feasible (λ, κ), a compact convex set (a linear image of a compact convex set), and $\overline{s}$ is the reach of the ray of w in it. □

Proposition 9.6 (Several contacts)**.** *Let K contacts act on the same rigid object, with independent measurements $y_k = \mathcal{M}^k(\nu_N^k)$ of their normal forces ($\mathcal{M}^k$ the operator of contact k), the same μ, and patches in a common plane (common coordinates, torques about the common normal),*

the out-of-plane wrench components being balanced without friction. The load w is carried if $w \in \mathcal{X}(\nu_N^1) \oplus \cdots \oplus \mathcal{X}(\nu_N^K)$, and the load factors of the grasp are

$$\boxed{\underline{s}_{\mathrm{grasp}}(w) = \inf_{q:\,\langle q,w\rangle=1} \sum_{k=1}^{K} \underline{h}(q; y_k, \mathcal{M}^k), \qquad \overline{s}_{\mathrm{grasp}}(w) = \inf_{q:\,\langle q,w\rangle=1} \sum_{k=1}^{K} \overline{h}(q; y_k, \mathcal{M}^k).} \tag{51}$$

Propositions 9.4 and 9.5 hold with the K contacts combined: summed objectives and a shared (a, b) for $\underline{s}_{\mathrm{grasp}}$, and, for $\overline{s}_{\mathrm{grasp}}$, a shared s with the corner forces of all contacts summed in the balance of $s\,w$. Moreover, $\underline{s}_{\mathrm{grasp}}$ is at least the load factor of $\bigoplus_k \mathcal{X}^-(y_k)$, the sum of the guaranteed sets of the individual contacts.

Proof. For fixed measures, $\bigoplus_k \mathcal{X}(\nu_N^k)$ is convex, compact, contains 0, and has support function $\sum_k h_{\mathcal{X}(\nu_N^k)}$ (Lemma 3.5), so the argument of Proposition 9.2(i) applies. The compatible family is a product of the individual families, so the infimum (resp. supremum) of the sum over compatible measures is the sum of the infima (resp. suprema). For the last claim, if $w_k \in \mathcal{X}^-(y_k)$ for each k, then $w_k \in \mathcal{X}(\nu_N^k)$ for every compatible ν_N^k, so $\sum_k w_k \in \bigoplus_k \mathcal{X}(\nu_N^k)$. □

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